Stefan Damjanovic

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Canadian and EU Citizen (Toronto, ON)

Education & Certificates

- MASc in Electrical/Computer Engineering from University of Toronto (Ongoing)
- BASc of Chemical Engineering (PEY stream) from Queen's University (Graduated)
- Queen Elizabeth II Graduate Scholarship in Science and Technology 2026-2027
- Medal In Chemical Engineering (Highest Cumulative GPA) 2020-2025

Work History/Projects/Internships

MASc Student, UofT Control Systems Group (Prof. Simpson-Porco), Toronto, ON 09/2025 – Present

- Researched stochastic control theory and Denoising Diffusion Probabilistic Models.
- Developed a distributionally-robust data-driven system identification framework that reduced mean and worst-case estimation errors by up to 85% and 95%, respectively.
- Investigated applications to financial portfolio optimization during black-swan events and training artificial intelligence models that generalize well to unseen data.
- Preprint available at https://arxiv.org/pdf/2610.09395
- Presented this work at the Physical AI Symposium 2026 in Osaka, Japan.

Director of Professional Development, University of Toronto, Toronto, ON 09/2026 – *Present*

- Organized resume-building and networking workshops, coordinating panel discussions tailored to graduate students with speakers from academia and industry.

Teaching Assistant, University of Toronto / Queen's University, 09/2021 – *Present*

- Responsible for grading all assessments from two Electrical Engineering courses (ECE537 Random Processes, and ECE410 Linear Control Systems).
- Independently led weekly tutorial sessions with +100 students and was selected to be an Exam Proctor for two Chemical Engineering courses (APSC 131, APSC 132).

Technical Mentor, UofT Machine Intelligence Student Team, Toronto, ON 09/2025 – 05/2026

- Mentored a team of 18 undergraduate engineering students in the development of emotionally adaptive artificial intelligence (AI) conversation agents.
- Integrated speech recognition, prosody analysis, and LLM dialogue models to give adaptive responses in real-time.

Founder/Developer, MyBodhi, Toronto, ON *05/2025 – 09/2025*

- Built a wellness app (MyBodhi) with an integrated AI personal development coach.
- Designed structured data models to adapt based on user practices, streaks, and lessons, enabling analytics and personalization.
- Implemented RAG-style context injection for improved question-answering using user data.

Researcher, Queen's University (NSERC USRA, Prof. Guay), Kingston, ON *05/2025 – 08/2025*

- Trained a neural network (gated recurrent unit) for accurate state recovery of nonlinear dynamical systems under severely-limited observations.
- Achieved successful recovery of full state for several chaotic systems (Lorenz, Chua, etc.)
- Target applications include robotic systems and chemical plants, where sensor data is limited due to cost constraints.

Control Systems Software Lead, Queen's Robotics Team, Kingston, ON *01/2025 – 08/2025*

- Designed the control systems for multiple robots to be capable of autonomously completing tasks, competing at the 2025 World Finals in Texas.
- Integrated model predictive control (MPC), resulting in autonomous real-time optimization of control sequences and obstacle avoidance.

Quality Operations/Validation Systems (QOVS) Intern, Sanofi, Toronto, ON *01/2023 – 12/2023*

- Prepared reports for Production and Quality teams to identify out-of-compliance issues.
- Created a *Smartsheet* metrics dashboard to justify $500,000 allocation towards QOVS.
- Optimized data collection schedules, reducing manual information collection time by 90%.

Research Student, Queen's University (NSERC USRA), Kingston, ON *05/2022 – 09/2022*

- Analyzed large genomics and protein-protein interaction datasets to investigate molecular pathogenesis pathways driving autoimmune diseases, with a primary focus on vitiligo.
- Implemented machine learning frameworks in Python and R to screen, model, and prioritize novel therapeutic molecule candidates for targeted intervention.